

Linear Algebra Notes: Matrices & Determinants

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1 Matrix Definitions and Basics (Page 1)

Definition

A matrix is a rectangular array of numbers or functions enclosed in brackets.

These numbers are called **entries** or **elements** of the matrix. The horizontal lines are called **rows** and the vertical lines are called **columns**.

If a matrix has the same number of rows and columns, it is called a **square matrix**.

Order 1: $[a]$

Solving system of variables $(x, y, z) : 5x + 3 = 13$

2 System Representations & Special Matrix Types (Page 2)

For a linear system of equations in x, y, z , if the entries are shown in matrix form, it is called a **coefficient matrix**.

Example: Find the coefficient matrix of the system:

$$\begin{cases} 5x - 2y + z = 0 \\ 3x + 4z = 0 \end{cases}$$

The coefficient matrix is:

$$\begin{bmatrix} 5 & -2 & 1 \\ 3 & 0 & 4 \end{bmatrix}$$

Row and Column Matrices

If a matrix has only one row or one column, it is called a **row matrix** or **column matrix** respectively.

Examples:

Row matrix: $a = [1 \ 2 \ 3]$

Column matrix: $b = \begin{bmatrix} 4 \\ 9 \\ 6 \end{bmatrix}$

Inner Product

The product of a row matrix and a column matrix is called the **inner product**:

$$a \cdot b = [a_1 \ a_2 \ \cdots \ a_n] \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = a_1b_1 + a_2b_2 + \cdots + a_nb_n$$

3 Transpose, Addition, and Equality of Matrices (Page 3)

Transpose of a Matrix

If $A = [a_{ij}]$ is a matrix of order $m \times n$, then the transpose of A , denoted A^T , is defined as:

$$A^T = [a_{ji}] \quad \text{of order } n \times m$$

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 9 & 8 \end{bmatrix} \implies A^T = \begin{bmatrix} 1 & 4 \\ 2 & 9 \\ 3 & 8 \end{bmatrix}$$

Addition of Matrices

If $A = [a_{jk}]$ and $B = [b_{jk}]$ are two matrices of the same size, then their sum $A + B = [a_{jk} + b_{jk}]$ is obtained by adding corresponding entries.

Example:

$$A = \begin{bmatrix} -4 & 6 & 3 \\ 0 & 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & -1 & 0 \\ 3 & 1 & 0 \end{bmatrix}$$
$$A + B = \begin{bmatrix} 1 & 5 & 3 \\ 3 & 2 & 2 \end{bmatrix}$$

Equality of Matrices

Two matrices are said to be equal if they are of the same order and their corresponding entries are equal:

$$a_{11} = b_{11}, \quad a_{12} = b_{12}, \quad \dots$$

4 Scalar and Matrix Multiplication (Page 4)

Scalar Multiplication

If A is a matrix of any order, then cA is a matrix obtained by multiplying the scalar c with each entry of the matrix.

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow 2A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

Matrix Multiplication

If $A = [a_{ik}]_{m \times n}$ and $B = [b_{jk}]_{r \times p}$ are two matrices, then the multiplication AB is defined if $r = n$ (i.e., number of columns in A = number of rows in B).

The resulting matrix $AB = [c_{jk}]_{m \times p}$, where:

$$c_{jk} = \sum_{l=1}^n a_{jl}b_{lk} = a_{j1}b_{1k} + a_{j2}b_{2k} + \cdots + a_{jn}b_{nk}$$

Example:

$$A = \begin{bmatrix} 4 & 3 \\ 7 & 2 \\ 9 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 5 \\ 1 & 6 \end{bmatrix}$$
$$AB = \begin{bmatrix} 4 \cdot 2 + 3 \cdot 1 & 4 \cdot 5 + 3 \cdot 6 \\ 7 \cdot 2 + 2 \cdot 1 & 7 \cdot 5 + 2 \cdot 6 \\ 9 \cdot 2 + 0 \cdot 1 & 9 \cdot 5 + 0 \cdot 6 \end{bmatrix} = \begin{bmatrix} 11 & 38 \\ 16 & 47 \\ 18 & 45 \end{bmatrix}$$

5 Types of Matrices (Page 5)

Triangular Matrices

- **Upper Triangular Matrix:** A square matrix that has non-zero entries only on and above the main diagonal, and all entries below the diagonal are zero.
- **Lower Triangular Matrix:** A square matrix with non-zero entries on and below the main diagonal, while all other entries above the diagonal are zero.

Diagonal Matrix

A square matrix that has non-zero entries only on the main diagonal, and any entry above or below the diagonal is zero, is called a **diagonal matrix**.

Identity Matrix

If all diagonal entries of a diagonal matrix are equal, it is called a **scalar matrix**. If all these entries are 1, it is called an **identity matrix**.

Symmetric Matrix

A square matrix A is symmetric if it is equal to its transpose:

$$A = A^T$$

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6 Skew-Symmetric, Orthogonal, and Complex Matrices (Page 6)

Skew-Symmetric Matrix

A matrix is skew-symmetric if it is equal to the negative of its transpose:

$$A = -A^T \quad \text{or} \quad A^T = -A$$

Example:

$$A = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -5 \\ -2 & 5 & 0 \end{bmatrix}$$

Orthogonal Matrix

A square matrix A is orthogonal if its transpose gives the inverse of A :

$$A^T = A^{-1} \implies A^T A = I$$

For a 2×2 orthogonal matrix $\begin{bmatrix} p & t \\ q & u \end{bmatrix}$:

$$p^2 + t^2 = 1, \quad q^2 + u^2 = 1, \quad pq + tu = 0$$

Rotation Matrix Example:

$$\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}, \quad \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Complex Matrix

Replacing a_{ij} with its complex conjugate $\bar{a}_{ij} = x - iy$, we get the **conjugate matrix** $\bar{A}$.

Taking the transpose of the conjugate matrix, $\bar{A}^T = [\bar{a}_{ji}]$, we get the **conjugate transpose** (or Hermitian transpose).

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7 Hermitian, Skew-Hermitian, and Unitary Matrices (Page 7)

- **Hermitian Matrix:** $\bar{A}^T = A$
- **Skew-Hermitian Matrix:** $\bar{A}^T = -A$
- **Unitary Matrix:** $\bar{A}^T = A^{-1} \implies A\bar{A}^T = I$

Hermitian Matrix Example

$$A = \begin{bmatrix} 0 & a - ib & c - id \\ a + ib & 1 & m - in \\ c + id & m + in & 2 \end{bmatrix}$$

Skew-Hermitian Matrix Example

$$A = \begin{bmatrix} i & 2+i \\ -2+i & 0 \end{bmatrix} \implies -A = \begin{bmatrix} -i & -2-i \\ 2-i & 0 \end{bmatrix} = \bar{A}^T$$

Unitary Matrix Example

$$A = \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$$
$$AA^* = \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1-i & 1+i \\ 1+i & 1-i \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} = I$$

8 Decomposition into Symmetric and Skew-Symmetric Parts (Page 8)

Show that any square matrix can be written as a sum of a symmetric and a skew-symmetric matrix:

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

Let $B = \frac{1}{2}(A + A^T) \implies B^T = \frac{1}{2}(A^T + A) = B$ (Symmetric).

Let $C = \frac{1}{2}(A - A^T) \implies C^T = \frac{1}{2}(A^T - A) = -C$ (Skew-Symmetric).

Example Problem: Express $A = \begin{bmatrix} 4 & 2 & -3 \\ 1 & 3 & -6 \\ -5 & 0 & -7 \end{bmatrix}$ as a sum of symmetric and skew-symmetric matrices.

Symmetric Part $\frac{1}{2}(A + A^T)$:

$$\frac{1}{2} \left(\begin{bmatrix} 4 & 2 & -3 \\ 1 & 3 & -6 \\ -5 & 0 & -7 \end{bmatrix} + \begin{bmatrix} 4 & 1 & -5 \\ 2 & 3 & 0 \\ -3 & -6 & -7 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} 8 & 3 & -8 \\ 3 & 6 & -6 \\ -8 & -6 & -14 \end{bmatrix} = \begin{bmatrix} 4 & 3/2 & -4 \\ 3/2 & 3 & -3 \\ -4 & -3 & -7 \end{bmatrix}$$

Skew-Symmetric Part $\frac{1}{2}(A - A^T)$:

$$\frac{1}{2} \left(\begin{bmatrix} 4 & 2 & -3 \\ 1 & 3 & -6 \\ -5 & 0 & -7 \end{bmatrix} - \begin{bmatrix} 4 & 1 & -5 \\ 2 & 3 & 0 \\ -3 & -6 & -7 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -6 \\ -2 & 6 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1/2 & 1 \\ -1/2 & 0 & -3 \\ -1 & 3 & 0 \end{bmatrix}$$

9 Determinants and Properties (Page 9)

A determinant is a number associated with an $n \times n$ matrix.

$$\text{Order 1: } |a| = a$$

$$\text{Order 2: } \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Properties of Determinants

1. If we interchange two rows/columns, then the value of the determinant changes sign.
2. If we add a constant multiple of a row/column to any other row/column, the determinant remains unchanged.
3. If we multiply a constant k to all the elements of a row/column, the value of the determinant changes by k times.
4. If all the elements of a row/column are zero, the value of the determinant is zero.
5. If any two rows/columns are identical, the value of the determinant is zero.
6. The determinant of a matrix and its transpose are the same.

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10 Additional Determinant Properties & Evaluation (Page 10)

Additional Properties

1. $\det(AB) = \det(A) \cdot \det(B)$
2. $\det(\bar{A}) = \overline{\det(A)}$

Theorem: The determinant of a triangular matrix is equal to the product of its diagonal elements.

$$\begin{vmatrix} -3 & 0 & 0 \\ 6 & 4 & 0 \\ -1 & 2 & 5 \end{vmatrix} = (-3) \cdot 4 \cdot 5 = -60$$

Example Evaluation

Find the value of the determinant by reducing into triangular form:

$$D = \begin{vmatrix} a & a & a \\ a & b & b \\ a & b & c \end{vmatrix}$$

Row Operations: $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$D = \begin{vmatrix} a & a & a \\ 0 & b-a & b-a \\ 0 & b-a & c-a \end{vmatrix}$$

Row Operation: $R_3 \rightarrow R_3 - R_2$:

$$D = \begin{vmatrix} a & a & a \\ 0 & b-a & b-a \\ 0 & 0 & c-b \end{vmatrix}$$

Since it is now an upper triangular matrix, the determinant is the product of the diagonal elements:

$$D = a(b-a)(c-b)$$

Determinant of an Orthogonal Matrix

The determinant of an orthogonal matrix is ± 1 .

Proof: Since A is an orthogonal matrix, we have:

$$A^T = A^{-1} \implies AA^{-1} = I$$

Taking the determinant on both sides:

$$\det(AA^{-1}) = \det(I) = 1$$

$$\det(A) \cdot \det(A^{-1}) = 1$$

Since $\det(A^{-1}) = \det(A^T)$:

$$\det(A) \cdot \det(A^T) = 1$$

Since $\det(A^T) = \det(A)$:

$$\det(A) \cdot \det(A) = 1 \implies (\det(A))^2 = 1$$

$$\implies \det(A) = \pm 1$$

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Determinant of a Unitary Matrix

The determinant of a unitary matrix has an absolute value of 1 ($|\det(A)| = 1$).

Proof: For a unitary matrix A :

$$\bar{A}^T = A^{-1} \implies AA^{-1} = I$$

Taking the determinant on both sides:

$$\det(AA^{-1}) = 1 \implies \det(A) \cdot \det(A^{-1}) = 1$$

$$\det(A) \cdot \det(\bar{A}^T) = 1$$

Since $\det(\bar{A}^T) = \det(\bar{A})$:

$$\det(A) \cdot \det(\bar{A}) = 1$$

Since $\det(\bar{A}) = \overline{\det(A)}$:

$$\det(A) \cdot \overline{\det(A)} = 1 \implies |\det(A)|^2 = 1$$

$$\implies |\det(A)| = 1$$

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11 Determinant Evaluation using Upper Triangular Form (IMG_0942)

Evaluate the matrix determinant by reducing it to upper triangular form:

$$|A| = \begin{vmatrix} 3 & 1 & -1 & 0 \\ 2 & 2 & 2 & 1 \\ -1 & 3 & 0 & 4 \\ 8 & 6 & -2 & 2 \end{vmatrix}$$

Step 1: Row operations to clear the first column

- $R_2 \rightarrow R_2 - \frac{2}{3}R_1$
- $R_3 \rightarrow R_3 + \frac{1}{3}R_1$
- $R_4 \rightarrow R_4 - \frac{8}{3}R_1$

$$|A| = \begin{vmatrix} 3 & 1 & -1 & 0 \\ 0 & \frac{4}{3} & \frac{8}{3} & 1 \\ 0 & \frac{10}{3} & -\frac{1}{3} & 4 \\ 0 & \frac{10}{3} & \frac{2}{3} & 2 \end{vmatrix}$$

Step 2: Row operations for second column

- $R_3 \rightarrow R_3 - \frac{10}{4}R_2$
- $R_4 \rightarrow R_4 - \frac{10}{4}R_2$

$$|A| = \begin{vmatrix} 3 & 1 & -1 & 0 \\ 0 & \frac{4}{3} & \frac{8}{3} & 1 \\ 0 & 0 & -\frac{21}{3} & \frac{6}{4} \\ 0 & 0 & -\frac{18}{3} & -\frac{2}{4} \end{vmatrix}$$

Step 3: Row operation for third column

- $R_4 \rightarrow R_4 - \frac{18}{21}R_3$

$$|A| = \begin{vmatrix} 3 & 1 & -1 & 0 \\ 0 & \frac{4}{3} & \frac{8}{3} & 1 \\ 0 & 0 & -\frac{21}{3} & \frac{6}{4} \\ 0 & 0 & 0 & -\frac{50}{28} \end{vmatrix}$$

Determinant Calculation: The determinant is the product of the main diagonal elements:

$$|A| = 3 \times \frac{4}{3} \times \left(-\frac{21}{3}\right) \times \left(-\frac{50}{28}\right) = 50$$

12 Systems of Equations, Linear Dependence, and Rank (IMG_0943)

Three Types of Solutions

For a system of linear equations:

1. Unique Solution Example:

$$\begin{cases} x + 2y = 5 \\ 2x + y = 4 \end{cases} \implies \text{Rank}(A) = 2, \text{Rank}([A|B]) = 2$$

2. Infinite Solutions Example:

$$\begin{cases} x + 2y = 5 \\ 2x + 4y = 10 \end{cases} \implies \text{Rank}(A) = 1, \text{Rank}([A|B]) = 1$$

3. No Solution Example:

$$\begin{cases} x + 2y = 5 \\ 2x + 4y = 7 \end{cases} \implies \text{Rank}(A) = 1 \neq \text{Rank}([A|B]) = 2$$

Linear Dependence (L.D.) Definition

A set of vectors $\{v_1, v_2, \dots, v_n\}$ are said to be ****Linearly Dependent (L.D.)**** if there exist scalars $r_1, r_2, \dots, r_n$ (not all zero) such that:

$$r_1v_1 + r_2v_2 + \dots + r_nv_n = 0$$

Otherwise, if all $r_i = 0$, the vectors are called ****Linearly Independent (L.I.)****.

Examples:

- $\{(1, 0), (2, 1)\}$ is Linearly Independent.
- $\{(1, 2, 3), (2, 4, 6)\}$ is Linearly Dependent.

Definition of Rank

The maximum number of Linearly Independent (L.I.) rows/columns in a matrix is called the **rank** of a matrix.

13 Elementary Operations & Echelon Form (IMG_0944)

Elementary Row Operations

The rank of a matrix remains unchanged by:

1. Interchange of two rows.
2. Multiplication of a row by a non-zero constant c .
3. Addition of a constant multiple of one row to another row.

Echelon Form

Reduction of a matrix to row-echelon form means:

- The first non-zero entry of the first row is non-zero.
- The first entry of the 2nd row will be zero below the leading entry of the 1st row.
- The number of leading zeros increases in each subsequent row.

Example Problem: Find the rank of:

$$A = \begin{bmatrix} 8 & -4 \\ -2 & 1 \\ 6 & -3 \end{bmatrix}$$

Applying row operations:

- $R_2 \rightarrow 4R_2 + R_1$
- $R_3 \rightarrow 4R_3 - 3R_1$

$$\begin{bmatrix} 8 & -4 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Since there is only 1 non-zero row, Rank = 1.

14 Solving Systems of Linear Equations via Gauss Elimination (IMG_0945 – IMG_0947)

Example 1: Unique Solution (IMG_0945)

Solve the following system using Gauss elimination:

$$\begin{cases} -x + y + 2z = 2 \\ 3x - y + z = 6 \\ -x + 3y + 4z = 4 \end{cases}$$

Augmented Matrix $[A|B]$:

$$\left[\begin{array}{ccc|c} -1 & 1 & 2 & 2 \\ 3 & -1 & 1 & 6 \\ -1 & 3 & 4 & 4 \end{array} \right]$$

Row Operations:

1. $R_2 \rightarrow R_2 + 3R_1$:

$$\left[\begin{array}{ccc|c} -1 & 1 & 2 & 2 \\ 0 & 2 & 7 & 12 \\ -1 & 3 & 4 & 4 \end{array} \right]$$

2. $R_3 \rightarrow R_3 - R_1$:

$$\left[\begin{array}{ccc|c} -1 & 1 & 2 & 2 \\ 0 & 2 & 7 & 12 \\ 0 & 2 & 2 & 2 \end{array} \right]$$

3. $R_3 \rightarrow R_3 - R_2$:

$$\left[\begin{array}{ccc|c} -1 & 1 & 2 & 2 \\ 0 & 2 & 7 & 12 \\ 0 & 0 & -5 & -10 \end{array} \right]$$

Back Substitution:

$$-5z = -10 \implies z = 2$$

$$2y + 7(2) = 12 \implies 2y = -2 \implies y = -1$$

$$-x + (-1) + 2(2) = 2 \implies -x + 3 = 2 \implies x = 1$$

Solution: $(x, y, z) = (1, -1, 2)$.

Example 2: Infinitely Many Solutions (IMG_0946)

Solve the system:

$$\begin{cases} -x + 2y - 5z = -21 \\ 4x - 8y + 3z = 16 \\ 3x - 6y + z = 7 \end{cases}$$

Augmented Matrix $[A|B]$:

$$\left[\begin{array}{ccc|c} -1 & 2 & -5 & -21 \\ 4 & -8 & 3 & 16 \\ 3 & -6 & 1 & 7 \end{array} \right]$$

Row Operations:

1. $R_2 \rightarrow R_2 + 4R_1$ and $R_3 \rightarrow R_3 + 3R_1$:

$$\left[\begin{array}{ccc|c} -1 & 2 & -5 & -21 \\ 0 & 0 & -17 & -68 \\ 0 & 0 & -14 & -56 \end{array} \right]$$

2. $R_3 \rightarrow R_3 - \frac{14}{17}R_2$:

$$\left[\begin{array}{ccc|c} -1 & 2 & -5 & -21 \\ 0 & 0 & -17 & -68 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Solving in terms of Parameter k :

$$-17z = -68 \implies z = 4$$

Let $x = k$:

$$-k + 2y - 5(4) = -21 \implies 2y = -21 + 20 + k = k - 1$$

$$y = \frac{k-1}{2}$$

Solution: $(x, y, z) = (k, \frac{k-1}{2}, 4)$ for any real parameter k .

Example 3: No Solution (IMG_0947)

Solve the system:

$$\begin{cases} 4x + y = 4 \\ 5x - 3y + z = 2 \\ -9x + 2y - z = 5 \end{cases}$$

Augmented Matrix $[A|B]$:

$$\left[\begin{array}{ccc|c} 4 & 1 & 0 & 4 \\ 5 & -3 & 1 & 2 \\ -9 & 2 & -1 & 5 \end{array} \right]$$

Row Operations:

1. $R_2 \rightarrow 4R_2 - 5R_1$ and $R_3 \rightarrow 4R_3 + 9R_1$:

$$\left[\begin{array}{ccc|c} 4 & 1 & 0 & 4 \\ 0 & -17 & 4 & -12 \\ 0 & 17 & -4 & 56 \end{array} \right]$$

2. $R_3 \rightarrow R_3 + R_2$:

$$\left[\begin{array}{ccc|c} 4 & 1 & 0 & 4 \\ 0 & -17 & 4 & -12 \\ 0 & 0 & 0 & 44 \end{array} \right]$$

The last row implies $0 = 44$, which is a contradiction ($0 \neq 44$).

Conclusion: No Solution (System is inconsistent).

15 Matrix Inversion Method (IMG_0948 – IMG_0949)

For a linear system $AX = B$, the solution is given by:

$$X = A^{-1}B$$

Using the augmented matrix approach $[A \mid I] \xrightarrow{\text{row operations}} [I \mid A^{-1}]$.

Problem: Solve the system using matrix inversion method:

$$\begin{cases} x + y + z = 3 \\ x + 2y + 3z = 4 \\ x + 4y + 9z = 6 \end{cases}$$

Set up Augmented Matrix $[A \mid I]$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 1 & 4 & 9 & 0 & 0 & 1 \end{array} \right]$$

Row Operations to find A^{-1} :

1. $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 3 & 8 & -1 & 0 & 1 \end{array} \right]$$

2. $R_1 \rightarrow R_1 - R_2$ and $R_3 \rightarrow R_3 - 3R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 2 & -1 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 2 & 2 & -3 & 1 \end{array} \right]$$

3. $R_3 \rightarrow \frac{1}{2}R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 2 & -1 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -3/2 & 1/2 \end{array} \right]$$

4. $R_1 \rightarrow R_1 + R_3$ and $R_2 \rightarrow R_2 - 2R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -5/2 & 1/2 \\ 0 & 1 & 0 & -3 & 4 & -1 \\ 0 & 0 & 1 & 1 & -3/2 & 1/2 \end{array} \right]$$

Thus, the inverse matrix is:

$$A^{-1} = \begin{bmatrix} 3 & -5/2 & 1/2 \\ -3 & 4 & -1 \\ 1 & -3/2 & 1/2 \end{bmatrix}$$

Computing $X = A^{-1}B$:

$$X = \begin{bmatrix} 3 & -5/2 & 1/2 \\ -3 & 4 & -1 \\ 1 & -3/2 & 1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 9 - 10 + 3 \\ -9 + 16 - 6 \\ 3 - 6 + 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Solution: $(x, y, z) = (2, 1, 0)$.

16 Matrix Inversion via Gauss-Jordan Elimination (IMG_0950)

Problem: Find the inverse of the matrix A using Gauss-Jordan elimination:

$$A = \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$$

Set up Augmented Matrix $[A \mid I]$:

$$\left[\begin{array}{ccc|ccc} -1 & 2 & 2 & 1 & 0 & 0 \\ 2 & -1 & 2 & 0 & 1 & 0 \\ 2 & 2 & -1 & 0 & 0 & 1 \end{array} \right]$$

Step-by-Step Row Operations:

1. $R_1 \rightarrow R_1 \cdot (-1)$:

$$\left[\begin{array}{ccc|ccc} 1 & -2 & -2 & -1 & 0 & 0 \\ 2 & -1 & 2 & 0 & 1 & 0 \\ 2 & 2 & -1 & 0 & 0 & 1 \end{array} \right]$$

2. $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 2R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & -2 & -2 & -1 & 0 & 0 \\ 0 & 3 & 6 & 2 & 1 & 0 \\ 0 & 6 & 3 & 2 & 0 & 1 \end{array} \right]$$

3. $R_2 \rightarrow \frac{1}{3}R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & -2 & -2 & -1 & 0 & 0 \\ 0 & 1 & 2 & 2/3 & 1/3 & 0 \\ 0 & 6 & 3 & 2 & 0 & 1 \end{array} \right]$$

4. $R_3 \rightarrow R_3 - 6R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & -2 & -2 & -1 & 0 & 0 \\ 0 & 1 & 2 & 2/3 & 1/3 & 0 \\ 0 & 0 & -9 & -2 & -2 & 1 \end{array} \right]$$

5. $R_3 \rightarrow R_3/(-9)$:

$$\left[\begin{array}{ccc|ccc} 1 & -2 & -2 & -1 & 0 & 0 \\ 0 & 1 & 2 & 2/3 & 1/3 & 0 \\ 0 & 0 & 1 & 2/9 & 2/9 & -1/9 \end{array} \right]$$

17 Gauss-Jordan Elimination Completed (IMG_0951)

Continuing the matrix inversion of $A = \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$ using augmented matrix $[A \mid I]$:

$$\left[\begin{array}{ccc|ccc} 1 & -2 & -2 & -1 & 0 & 0 \\ 0 & 1 & 2 & 2/3 & 1/3 & 0 \\ 0 & 0 & 1 & 2/9 & 2/9 & -1/9 \end{array} \right]$$

Row Operations to clear column 3:

- $R_2 \rightarrow R_2 - 2R_3$
- $R_1 \rightarrow R_1 + 2R_3$

$$\left[\begin{array}{ccc|ccc} 1 & -2 & 0 & -5/9 & 4/9 & -2/9 \\ 0 & 1 & 0 & 2/9 & -1/9 & 2/9 \\ 0 & 0 & 1 & 2/9 & 2/9 & -1/9 \end{array} \right]$$

Row Operation to clear column 2:

- $R_1 \rightarrow R_1 + 2R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1/9 & 2/9 & 2/9 \\ 0 & 1 & 0 & 2/9 & -1/9 & 2/9 \\ 0 & 0 & 1 & 2/9 & 2/9 & -1/9 \end{array} \right]$$

Inverse Matrix Result:

$$A^{-1} = \frac{1}{9} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$$

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18 LU Decomposition Theory (IMG_0952)

For a system of linear equations $AX = b$, assume $A = LU$, where:

- L is a **lower triangular matrix**.
- U is an **upper triangular matrix**.

The system becomes:

$$(LU)X = b \implies L(UX) = b$$

Let $UX = Y$, then the system splits into two triangular steps:

1. Solve $LY = b$ for Y using **forward substitution**.
2. Solve $UX = Y$ for X using **back substitution**.

2×2 Matrix Example Form:

$$\begin{bmatrix} a & 0 \\ b & c \end{bmatrix} \begin{bmatrix} d & f \\ 0 & e \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$
$$\begin{bmatrix} ad & af \\ bd & bf + ce \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Types of LU Decomposition Methods

1. **Doolittle's Method:** All diagonal elements of the lower triangular matrix L are set to unity ($l_{ii} = 1$).
2. **Crout's Method:** All diagonal elements of the upper triangular matrix U are set to unity ($u_{ii} = 1$).

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19 LU Factorization Example (IMG_0953)

Solve the linear system using Doolittle's LU Decomposition:

$$\begin{cases} x + y + 2z = -1 \\ 4x + 5y + 13z = -7 \\ 10x + 14y + 43z = -25 \end{cases}$$

Matrix Factorization Setup ($A = LU$):

$$\begin{bmatrix} 1 & 1 & 2 \\ 4 & 5 & 13 \\ 10 & 14 & 43 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Multiplying L and U :

$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

Equating Elements to Solve for l_{ij} and u_{ij} :

$$u_{11} = 1, \quad u_{12} = 1, \quad u_{13} = 2$$

$$l_{21}u_{11} = 4 \implies l_{21} = 4$$

$$l_{21}u_{12} + u_{22} = 5 \implies 4(1) + u_{22} = 5 \implies u_{22} = 1$$

$$l_{21}u_{13} + u_{23} = 13 \implies 4(2) + u_{23} = 13 \implies u_{23} = 5$$

$$l_{31}u_{11} = 10 \implies l_{31} = 10$$

$$l_{31}u_{12} + l_{32}u_{22} = 14 \implies 10(1) + l_{32}(1) = 14 \implies l_{32} = 4$$

$$l_{31}u_{13} + l_{32}u_{23} + u_{33} = 43 \implies 10(2) + 4(5) + u_{33} = 43 \implies u_{33} = 3$$

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20 Forward & Back Substitution (IMG_0955)

From the computed factors:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 10 & 4 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 5 \\ 0 & 0 & 3 \end{bmatrix}$$

Step 1: Solve $LY = b$

$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 10 & 4 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} -1 \\ -7 \\ -25 \end{bmatrix}$$

Using forward substitution:

$$y_1 = -1$$

$$4y_1 + y_2 = -7 \implies 4(-1) + y_2 = -7 \implies y_2 = -3$$

$$10y_1 + 4y_2 + y_3 = -25 \implies 10(-1) + 4(-3) + y_3 = -25 \implies y_3 = -3$$

Thus, $Y = \begin{bmatrix} -1 \\ -3 \\ -3 \end{bmatrix}$.

Step 2: Solve $UX = Y$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 5 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \\ -3 \end{bmatrix}$$

Using back substitution:

$$\begin{aligned} 3x_3 &= -3 \implies x_3 = -1 \\ x_2 + 5x_3 &= -3 \implies x_2 + 5(-1) = -3 \implies x_2 = 2 \\ x_1 + x_2 + 2x_3 &= -1 \implies x_1 + 2 + 2(-1) = -1 \implies x_1 = -1 \end{aligned}$$

Final Solution: $(x_1, x_2, x_3) = (-1, 2, -1)$.

21 LU Decomposition Computation Example (IMG_0956)

Compute the LU decomposition of matrix A using Doolittle's method:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Setup:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Calculations:

- First row of U : $u_{11} = 1, u_{12} = 1, u_{13} = 1$
- First column of L : $l_{21} = 1, l_{31} = 1$
- Second row of U : $u_{22} = 2 - 1(1) = 1, u_{23} = 2 - 1(1) = 1$
- Second column of L : $l_{32} = 2 - 1(1) = 1$
- Third row element of U : $u_{33} = 3 - (1(1) + 1(1)) = 1$

Resulting LU Factorization:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$